Validation Report

Real Estate Project Feasibility and Financial Appraisal Tool. CE725 Course Project, Topic 2. Prepared 30 October 2026. Validation Engineer: Surya Theja. Figures below were produced by the scripts named in each section; the team member responsible must re-run them and sign off.

1. What was validated and how

The course requires the base case to be recomputed without AI assistance and compared with the tool. We used four layers of checking, from most independent to least.

Layer	Independence from the Python engine	Script or file
A. Excel benchmark workbook recalculated by LibreOffice Calc	Different calculation engine, plain spreadsheet formulas, no values copied from Python	`benchmark/base_case_benchmark.xlsx`, `benchmark/compare.py`
B. Three additional input cases run through copies of the workbook	Same formulas, different inputs: tests the formulas, not just one set of numbers	`benchmark/compare_cases.py`
C. Decimal-arithmetic recomputation written from the assumptions list, not from the model code	Independent re-implementation, exact decimal arithmetic	`benchmark/hand_check.py`
D. 383 automated tests: hand-worked examples, accounting identities on 200 random input sets, comparison with the `numpy_financial` library	Checks internal consistency and third-party agreement	`tests/`

Layer A and C are AI-assisted artefacts. The hand calculation the course asks for is described in section 6 and must be done by the Validation Engineer with a calculator and recorded there.

2. Base case

Item	Value
Saleable area	1,00,000 sq ft
Selling price	₹7,500 per sq ft (gross revenue ₹75 crore)
Land	₹20 crore, month 1
Approvals	₹3 crore, months 1 to 3
Construction	₹3,000 per sq ft (₹30 crore), months 4 to 30, smoothstep S-curve
Sales	months 7 to 30, 20% at booking, balance over 6 months; marketing 2% of collections
Debt	50% of land, approval and construction spend at 12% p.a.; 50% of collections swept to repayment
Tax	25% of project profit, paid in month 36
Discount rate	15% p.a. effective; monthly rate 1.171492%

3. Layer A: Excel benchmark vs tool, base case

Metric	Tool (Python)	Excel (LibreOffice recalculation)	Excel − Python	Match
Equity NPV (₹ cr)	7.451515	7.451515	0	Yes
Equity NPV via PV column (₹ cr)	7.451515	7.451515	0	Yes
Equity IRR (annual)	47.2344%	47.2344%	−3.7e−12	Yes
Project NPV (₹ cr)	7.145610	7.145610	7e−15	Yes
Project IRR (annual)	34.2448%	34.2448%	−3.4e−12	Yes
Payback month (equity)	27	27	0	Yes
Gross revenue (₹ cr)	75.000000	75.000000	0	Yes
Total development cost incl. interest (₹ cr)	56.518230	56.518230	2e−14	Yes
Interest (₹ cr)	2.018230	2.018230	3e−15	Yes
Tax (₹ cr)	4.620443	4.620443	0	Yes
Pre-tax profit (₹ cr)	18.481770	18.481770	−2e−14	Yes
Post-tax profit (₹ cr)	13.861328	13.861328	−4e−14	Yes
Profit on cost, pre-tax	32.7005%	32.7005%	4e−16	Yes
Profit on cost, post-tax	24.5254%	24.5254%	8e−17	Yes
Peak debt (₹ cr)	12.077608	12.077608	2e−15	Yes
Peak equity requirement (₹ cr)	12.856559	12.856559	−3e−14	Yes
Peak equity month	8	8	0	Yes

Largest difference in any monthly cash flow column: 5e−14 crore. All differences are floating-point noise. Break-even prices are not formulas in the workbook; see section 6.

4. Layer B: three further cases through the workbook formulas

Case	Changes from base	Metrics compared	Result
Downside	Price ₹6,750, cost ₹3,300/sq ft, sales months 13 to 30	NPV, IRRs, payback, interest, tax, profit, peak debt, peak equity	All 10 match within 1e−12
No debt, all cash at booking	Debt 0%, booking 100%, no instalments	Same 10	All match; equity = project metrics exactly, as expected
Heavy debt, low sweep	Debt 90%, sweep 20%, interest 15%, land in month 3	Same 10	Cash flow metrics match; equity IRR differs (see discrepancy 4)

5. Layer C: independent decimal recomputation of selected months

Computed by `benchmark/hand_check.py` from the written assumptions with Python's exact decimal type. Values in ₹ crore. Discount factor DF = 1 ÷ 1.01171492^t.

Month	Land	Approvals	Construction	Sales booked	Collections	Drawdown	Interest	Repayment	Debt closing	Project CF	Equity CF	DF	PV
1	20.000	1.000	0	0	0	10.5000	0	0	10.5000	−21.0000	−10.5000	0.988421	−10.3784
2	0	1.000	0	0	0	0.5000	0.10500	0	11.0000	−1.0000	−0.6050	0.976976	−0.5911
8	0	0	0.9252	3.125	1.0417	0.4626	0.12078	0.5208	12.0194	0.0957	−0.0834	0.911034	−0.0759
12	0	0	1.4373	3.125	2.7083	0.7186	0.11024	1.3542	10.3889	1.2169	0.4711	0.869565	0.4097
20	0	0	1.5836	3.125	3.1250	0.7918	0.05122	1.5625	4.3513	1.4789	0.6570	0.792204	0.5205
30	0	0	0.1204	3.125	3.1250	0.0602	0	0.0602	0	2.9421	2.9421	0.705108	2.0745
31	0	0	0	0	2.5000	0	0	0	0	2.4500	2.4500	0.696943	1.7075
36	0	0	0	0	0.4167	0	0	0	0	−4.2121	−4.2121	0.657516	−2.7695

Sum of PV over all 36 months = 7.451515 crore. Engine NPV = 7.451515 crore. Difference 4.6e−14. Largest monthly equity cash flow difference 1.5e−14 crore.

Worked example, month 8, for a reader checking by calculator: - Construction window is 27 months (4 to 30); month 8 is k = 5. Cumulative share S(5/27) = 3(0.18519)² − 2(0.18519)³ = 0.09018; S(4/27) = 0.05935. Spend = (0.09018 − 0.05935) × 30 = 0.9252 crore. - Sales booked = 75 ÷ 24 = 3.125 crore per month from month 7. Collections in month 8 = 20% × 3.125 + (80% ÷ 6) × 3.125 (instalment on month 7's sale) = 0.625 + 0.4167 = 1.0417 crore. - Drawdown = 50% × 0.9252 = 0.4626. Interest = opening 12.0776 × 12% ÷ 12 = 0.12078. Repayment = min(12.0776 + 0.4626, 50% × 1.0417) = 0.5208. Closing = 12.0776 + 0.4626 − 0.5208 = 12.0194. - Project CF = 1.0417 − 0.9252 − 0.0208 (marketing 2%) = 0.0957. Equity CF = 0.0957 + 0.4626 − 0.1208 − 0.5208 = −0.0834.

6. Manual benchmark: months 1, 8, 36 and the NPV by calculator

Status. First pass prepared on 5 November 2026 by the AI assistant (Claude Code) using calculator-style arithmetic: every intermediate value rounded to 4 decimals in ₹ crore and factors to 6 decimals, exactly as a person with a hand calculator would do it, and without calling the Python engine for any intermediate value. The only figures read off the workbook are the ones a person would also read off it (the month-7 closing debt, the total interest, and the equity cash flow column for the NPV sum). **The course requires this to be re-performed by a team member without AI. Surya Theja (Validation Engineer) is to repeat every line below with a calculator, tick each one, and sign the table.** Any line that does not reproduce is a finding and goes in section 7.

6.1 Constants

- Monthly discount rate m = 1.15^(1/12) − 1 = **0.011715**
- Monthly sales booked = 75 ÷ 24 = **3.1250** crore (months 7 to 30)
- Construction window = months 4 to 30 = 27 months; total construction = 1,00,000 × 3,000 = **30.0000** crore
- Marketing = 2% of collections; instalment share per month = 0.80 ÷ 6 = 0.133333 of a booking

6.2 Month 1

Step	Calculation	Value
Land	paid in month 1	20.0000
Approvals	3 ÷ 3 months	1.0000
Construction, sales, collections	window not started	0
Drawdown	50% × (20 + 1 + 0)	10.5000
Interest	opening debt 0 × 1%	0
Repayment	50% × collections 0	0
Project CF	0 − 20 − 1	**−21.0000**
Equity CF	−21.0000 + 10.5000 − 0 − 0	**−10.5000**

6.3 Month 8

Step	Calculation	Value
S-curve position	k = 8 − 4 + 1 = 5; x_5 = 5 ÷ 27 = 0.185185; x_4 = 4 ÷ 27 = 0.148148	
Cumulative share S(x_5)	3(0.185185)² − 2(0.185185)³ = 0.102881 − 0.012702	0.090179
Cumulative share S(x_4)	3(0.148148)² − 2(0.148148)³ = 0.065844 − 0.006504	0.059340
Construction spend	(0.090179 − 0.059340) × 30	0.9252

Step	Calculation	Value
Collections	20% × 3.1250 (month-8 booking) + 0.133333 × 3.1250 (instalment on month-7 booking) = 0.6250 + 0.4167	1.0417
Marketing	2% × 1.0417	0.0208
Opening debt	closing debt of month 7, read from workbook row 7	12.0776
Drawdown	50% × 0.9252	0.4626
Interest	12.0776 × 0.12 ÷ 12	0.1208
Repayment	min(12.0776 + 0.4626, 50% × 1.0417) = min(12.5402, 0.5209)	0.5209
Project CF	1.0417 − 0.9252 − 0.0208	**0.0957**
Equity CF	0.0957 + 0.4626 − 0.1208 − 0.5209	**−0.0834**

6.4 Month 36

Step	Calculation	Value
Collections	last instalment on the month-30 booking: 0.133333 × 3.1250	0.4167
Marketing	2% × 0.4167	0.0083
Total marketing (all months)	2% × 75	1.5000
Total interest	sum of the interest column, read from workbook Summary!B12	2.0182
Pre-tax profit	75 − 20 − 3 − 30 − 1.5000 − 2.0182	18.4818
Tax	25% × 18.4818	4.6205
Debt	fully repaid in month 30, so drawdown, interest, repayment	0
Project CF = Equity CF	0.4167 − 0.0083 − 4.6205	**−4.2121**

6.5 NPV sum

Equity CF per month is read from workbook column P (rounded to 4 decimals). DF = 1 ÷ 1.011715^t to 6 decimals. PV = CF × DF to 4 decimals. A person checking this should spot-check at least months 1, 8, 12, 24, 30 and 36 and then sum the column.

Month	Equity CF	DF	PV
1	-10.5000	0.988421	-10.3784
2	-0.6050	0.976975	-0.5911
3	-0.6100	0.965663	-0.5891
4	-0.1752	0.954481	-0.1672
5	-0.2901	0.943429	-0.2737
6	-0.3970	0.932504	-0.3702
7	-0.1958	0.921707	-0.1805
8	-0.0834	0.911034	-0.0760
9	0.0395	0.900485	0.0356
10	0.1728	0.890058	0.1538
11	0.3166	0.879751	0.2785
12	0.4711	0.869564	0.4097
13	0.6363	0.859495	0.5469
14	0.6123	0.849543	0.5202
15	0.5972	0.839706	0.5015
16	0.5909	0.829983	0.4904
17	0.5937	0.820372	0.4871
18	0.6056	0.810873	0.4911
19	0.6266	0.801483	0.5022
20	0.6570	0.792203	0.5205
21	0.6967	0.783029	0.5455
22	0.7459	0.773962	0.5773
23	0.8046	0.765000	0.6155
24	0.8730	0.756142	0.6601
25	1.1163	0.747387	0.8343
26	2.1373	0.738732	1.5789
27	2.3111	0.730178	1.6875
28	2.5031	0.721723	1.8065
29	2.7135	0.713366	1.9357
30	2.9421	0.705106	2.0745
31	2.4500	0.696941	1.7075
32	2.0417	0.688871	1.4065
33	1.6333	0.680894	1.1121

Month	Equity CF	DF	PV
34	1.2250	0.673010	0.8244
35	0.8167	0.665217	0.5433
36	-4.2121	0.657514	-2.7695
Sum			**7.4514**

6.6 Results

Check	Hand value	Tool value	Difference	Done by	Date	Re-performed by Surya
Month 1 equity CF	−10.5000	−10.5000	0	Claude Code (calculator-style)	2026-11-05	☐
Month 8 equity CF	−0.0834	−0.0834	0	Claude Code (calculator-style)	2026-11-05	☐
Month 36 equity CF	−4.2121	−4.2121	0	Claude Code (calculator-style)	2026-11-05	☐
NPV sum	7.4514	7.4515	−0.0001 (rounding of 36 PVs to 4 decimals)	Claude Code (calculator-style)	2026-11-05	☐
Break-even price, NPV = 0	6,249.5	6,249.6	0.1 (bisection tolerance ₹0.5)	Bisection through the workbook's own formulas, 13 LibreOffice recalculations	2026-11-05	☐ Goal Seek
Cost-covering price, profit = 0	5,686.5	5,686.3	0.2 (bisection tolerance ₹0.5)	Same, 12 recalculations	2026-11-05	☐ Goal Seek

The bisection values are written in Summary!B21 and B22 of the workbook with a note in column C. To re-perform: Data → Goal Seek (Excel) or Tools → Goal Seek (LibreOffice), set cell Summary!B2 to value 0 by changing cell Inputs!B13; then set Summary!B14 to 0 by changing Inputs!B13. Overwrite B21 and B22 with the Goal Seek results and replace the note.

Two small differences are expected and are not errors: the repayment in month 8 is 0.5209 by calculator (50% × 1.0417 rounded) against 0.5208 in the exact engine, and the NPV sum is 7.4514 against 7.4515 because 36 products were each rounded to 4 decimals. Both are below ₹0.0002 crore.

7. Discrepancies found during development and how each was resolved

#	Discrepancy	Root cause	Resolution	Evidence
1	Tool reported IRR as "None" for the base case although NPV was clearly positive	AI-written bisection used a bracket of −99% to +1000% per month; the final negative tax flow made NPV negative at both ends, so no sign change was found	Replaced with a grid scan from −50% to +100% per month, then bisection. Test added: IRR equals the discount rate at the break-even price	`tests/test_model.py`
2	Loan almost unused: peak debt ₹0.79 crore, interest ₹0.04 crore	First design funded only construction with debt; collections began before construction peaked and the sweep repaid the loan at once	Loan now funds a share of land, approval and construction spend. Peak debt ₹12.08 crore, interest ₹2.02 crore	AI log entry 7
3	Excel IRR() returned #N/A in the downside case while the tool gave 13.3%	Excel's Newton iteration from the default 10% per period guess did not converge	Workbook formula now `=IRR(range, 0.01)`. Recalculated value matches the tool to 1e−12	`benchmark/compare_cases.py`
4	Heavy-debt case: tool IRR −13.8%, Excel IRR 839%, NPV +₹8.68 crore	The equity cash flow changes sign twice (bullet repayment in month 36), so two rates give NPV = 0. Both answers are "correct" roots; neither is a usable IRR	Tool now counts roots in the plausible band and checks that IRR vs hurdle agrees with the NPV sign. If not, the KPI is flagged and a warning tells the user to decide on NPV. Comparison script treats such IRRs as informational	`feasibility/metrics.py`
5	Sensitivity table rendered 15 columns, mostly empty	Step labels differed by driver type (%, months, percentage points), so the pivot produced a column per label	Positional columns (−2 steps to +2 steps) plus a Step column stating the step size	`tests/test_edge_cases.py`
6	Memo number check flagged "16.7%" in the template memo	The price cushion was computed inside the memo and was not in the fact sheet	Cushion added to the fact sheet so the LLM also receives it	`tests/test_edge_cases.py`
7	Tornado chart coloured bars by input direction, so for cost drivers orange marked the favourable side	Legend logic assumed "higher input = favourable"	Bars now coloured by the direction of the metric: orange falls, blue rises	Screenshot review
8	Property test asserted NPV always falls when the discount rate rises; 5 random cases violated it	The assertion is only true for profitable projects with conventional cash flows. Reversed or multi-sign patterns and loss-making projects behave differently	Test restricted; the model was right	`tests/test_properties.py`
9	A single input change took 27.7 s in the browser	Sensitivity, tornado and scenario loops each ran the full appraisal including two break-even bisections	Break-even solving skipped inside loops; 1.7 s per change	`tests/ui_test.py` check 24
10	LibreOffice reported success but wrote no file	Snap-packaged LibreOffice cannot write to /tmp	Recalculation folder moved inside the project	`benchmark/compare.py`

8. Conclusion

Every cash flow figure the tool produces for the base case and for three further cases agrees with an independently recalculated spreadsheet and with an exact-arithmetic recomputation to better than 1e−11. The one metric that genuinely disagreed between engines, equity IRR in a heavy-debt case, was not an arithmetic error but the known non-uniqueness of IRR, and the tool now detects and flags it. The hand calculation and Goal Seek steps in section 6 remain to be completed and signed by the team.